Improving Neck and Jawline Aesthetics With OnabotulinumtoxinA by Minimizing Platysma Muscle Contraction Effects: Efficacy and Safety Results in a Phase 3 Randomized, Placebo-Controlled Study

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Abstract

Background: Platysma prominence (PP) describes the noticeable appearance of the platysma muscle upon contraction, causing a less defined jawline contour and vertical neck bands.

Objectives: The objective of this study was to assess the safety and efficacy of onabotulinumtoxinA for improvement of PP in adults.

Methods: Participants with moderate to severe (Grade 3 to 4) PP at maximum contraction received onabotulinumtoxinA or placebo on Day 1 and were monitored for 120 days. OnabotulinumtoxinA dosage (26, 31, or 36 U) was customized based on baseline PP severity on each side of the neck.

Results: Efficacy analyses were conducted in the intent-to-treat (ITT) population (all randomized participants), and modified ITT population (mITT; psychosocially impacted by PP appearance). Results from ITT and mITT populations were comparable. As assessed by investigators, 76.7% of onabotulinumtoxinA mITT participants achieved ≥1-grade improvement vs 21.2% in the placebo group, and 41.0% vs 2.2% ($P < .0001$) achieved ≥2-grade improvement at Day 14. As assessed by participants, 79.9% of onabotulinumtoxinA mITT participants vs 21.8% in the placebo group and 40.8% vs 3.9% ($P < .0001$) achieved ≥1- or ≥2-grade improvement, respectively, at Day 14. OnabotulinumtoxinA responder rates remained higher than placebo through Day 120, gradually declining over time. OnabotulinumtoxinA participants reported significantly higher satisfaction with treatment effect, less bother from jawline and vertical neck bands, and lower psychosocial impact from PP than placebo at Day 14 ($P < .0001$). OnabotulinumtoxinA effectively improved self-perceived jawline definition and was well tolerated.

Conclusions: OnabotulinumtoxinA was well tolerated and effective at improving moderate to severe PP, including neck bands and jawline definition.

Level of Evidence: 1 (Therapeutic)

A well-defined lower face and neck signal a youthful facial appearance.[1,2] The platysma muscle plays a crucial role in lower facial and neck aesthetics by smoothly draping the area from the upper thoracic region to the lower face and above the mandible.[3-5] However, as the platysma muscle ages, it can contribute to a less defined jawline, accentuating jowling and visible vertical neck bands.[3,6,7] This condition is referred to as platysma prominence (PP) and is commonly associated with aging, although it can also affect younger individuals.[8]

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PP presentation can vary anatomically within the same individual, leading to an asymmetrical presentation of PP.[9]

PP is caused by the contraction of the platysma muscle over time, which leads to shortening of the muscle fibers and may result in a more pronounced appearance.[6,10] As a result, the platysma muscle may contract noticeably during normal activities like smiling or talking, becoming visible and potentially bothersome in appearance.[11,12] Other age-related changes, such as cervical skin laxity, platysma muscle diastasis, or cervical fat loss, can further exacerbate PP.[3] The changes in the lower face and neck contours caused by PP can negatively impact an individual's psychological well-being and self-esteem.[7,13] Moreover, PP can interfere with facial expressions, resulting in an unattractive appearance that does not accurately portray the person's emotional state.[14]

OnabotulinumtoxinA (onabotA; BOTOX Cosmetic, Vistabel, Allergan Aesthetics, an AbbVie company, Irvine, CA) demonstrated efficacy in improving moderate-to-severe PP at various doses (26, 31, and 36 U, or 52, 62, and 72 U) in a recent phase 2 dose-escalating study, with the 26, 31, and 36 U dose group showing a more favorable safety profile.[15] Additionally, a pivotal phase 3 study conducted in the United States and Canada confirmed the safety of onabotA dosages of 26, 31, and 36 U and showed significant and clinically meaningful improvements in PP.[16] In this phase 3 study, we evaluated the efficacy and safety of onabotA (26, 31, and 36 U) compared to placebo in participants from multiple countries, consisting of the United Kingdom, Germany, Belgium, the United States, and Canada.

METHODS

Study Design

This was a 4-month, multicenter, randomized, double-blind, placebo-controlled Phase 3 study conducted from August 2021 to June 2023. Eligible participants were randomized on Day 1 in a 1:1 ratio to receive onabotA or placebo, with central by block stratified randomization. Randomization was stratified by investigational site and baseline severity on Day 1. The study included 7 visits: screening (Day −14 to Day −7), randomization/treatment administration/baseline visit (Day 1), follow-up visits (Days 14, 30, 60, 90), and study exit (Day 120). On Day 1, the study drug was injected into the platysma muscle. The reconstitution of the study drug was performed by an unblinded designated staff member with no role in other study procedures. The investigator, study site personnel, and participant remained blinded to treatment throughout the study.

The study received approval from an independent ethical committee and an institutional review board (Advarra, Columbia, MD) and adhered to the guidelines of the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) and the Declaration of Helsinki. Participants provided their written informed consent before any study-related screening procedures (Clinicaltrials.gov ID, NCT04994535; EudraCT ID, 2021-000240-22).

Participants

Eligible participants were 18 years or older, with a BMI between 18 kg/m^2 and 30 kg/m^2, and PP Grade 3 (Moderate) or 4 (Severe) on both the left and the right sides at maximum contraction. Platysma prominence severities on the left and right sides did not have to be identical. On Day 1, PP severity was evaluated independently by the investigator with the Clinician Allergan Platysma Prominence Scale (C-APPS) and by the participants with the Participant Allergan Platysma Prominence Scale (P-APPS). Additionally, on Day 1, participants had to have responses of *Somewhat bothered*, *A lot bothered*, or *Extremely bothered* by the appearance of their jawline (Bother Assessment Scale–Platysma Prominence [BAS-PP] Item 2) and their vertical neck bands (BAS-PP Item 1).

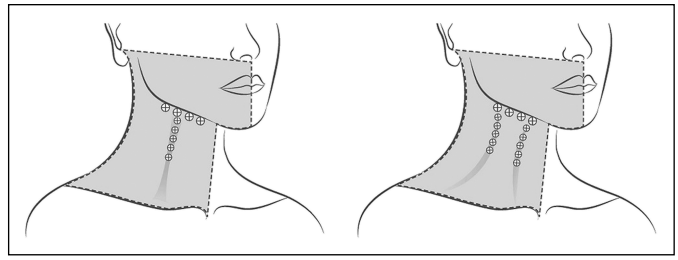


Figure 1. Injection pattern diagram. The number of bands treated per side depends on the severity of the condition. Diagram initially published by Rohrich et al.[15] Used with permission from Wolters Kluwer Health, Inc. (Waltham, MA).

Participants who had excessively loose skin in the lower face, neck, or décolletage area, submental fat, jowls, and/or medical conditions that might` pose an increased risk with onabotA exposure or that could interfere with study assessments were excluded from the study. Individuals with a known immunization or hypersensitivity to any botulinum toxin serotype were also excluded.

Treatment

OnabotA or placebo was administered as superficial intramuscular injections, considering the distinct anatomical structure and location of this mimetic muscle. To ensure superficial administration, trained aesthetic investigators pinched and lifted the skin along with the underlying platysma muscle to isolate it from adjacent neck structures during injection. Depending on the severity of the condition at baseline, participants received an individually adjusted injection scheme (Figure 1). Participants were given a fixed dose of either onabotA 16 U (8 U per side) or placebo along the jawline in the upper segment of the platysma muscle, specifically at the transition line of pars cervicalis to pars facialis, approximately 1 to 2 cm below and parallel to the mandibular border. Four jawline injections were given on each side (0.05 mL each), spaced 1 to 2 cm apart. Additionally, participants received onabotA 5 U or placebo per vertical neck band, injected into a total of 2 to 4 continuous vertical bands (1 or 2 bands per side). Each treated band received 5 injections (0.025 mL each), spaced 1 to 2 cm apart. Consequently, participants could receive a total onabotA dose of 26, 31, or 36 U for the jawline and neck.

Efficacy and Safety Outcome Measures

Efficacy analyses were conducted in 2 populations: the intent-to-treat (ITT) population included all randomized participants, whereas the modified ITT (mITT) population consisted of all randomized participants who also had a baseline summary score of ≥19 on the Appearance of Neck and Lower Face Questionnaire (ANLFQ): Impacts, measuring psychosocial impact related to PP appearance before treatment.

The study incorporated a composite primary endpoint and 2 coprimary endpoints to meet global regulatory requirements. The composite primary endpoint, analyzed in the ITT population, was the achievement of a Grade 1 or 2 (Minimal or Mild) and a ≥2-grade improvement from baseline, as assessed by both the investigator with the C-APPS and the participant with the P-APPS at maximum contraction on Day 14. The 2 coprimary efficacy endpoints analyzed in the mITT population were achievement of investigator-assessed and participant-assessed ≥2-grade improvement from baseline with the

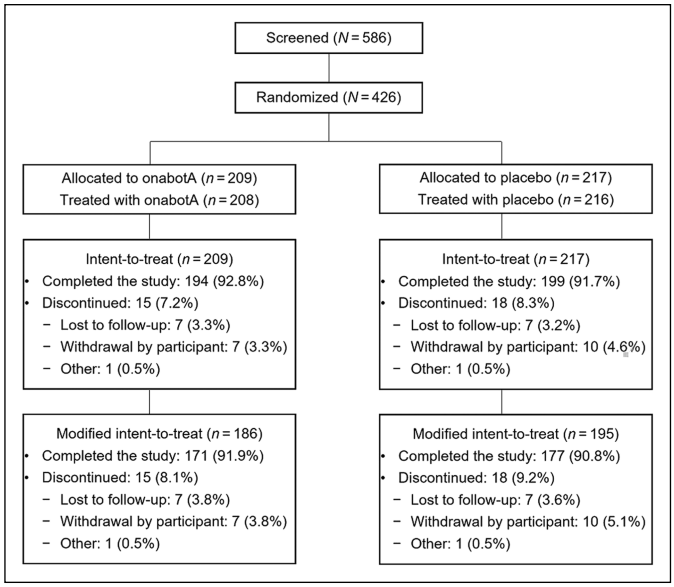


Figure 2. Participant disposition. Percentages are based on the population sample size within each box. OnabotA, onabotulinumtoxinA.

C-APPS and P-APPS, respectively, at maximum contraction on Day 14. The C-APPS and P-APPS are validated photonumeric scales.[15,17]

Secondary efficacy endpoints included achievement of investigator- and participant-assessed Grade 1 or 2 at maximum contraction over time (ITT), achievement of Grade 1 or 2 as assessed by the participant with the P-APPS at maximum contraction on Day 14 (mITT), participant satisfaction with the effect of treatment on Day 14 on the ANLFQ: Satisfaction (Follow-up) Item 5 (ITT and mITT), participant bother with the jawline and vertical neck bands appearance on Day 14 by BAS-PP Items 2 and 1, respectively (ITT and mITT), change in the ANLFQ: Impacts summary score from baseline to Day 14 (ITT), as well as to Days 30, 60, and 90 (mITT), and achievement of ≥1-grade improvement at maximum contraction over time based on the investigator's and participant's assessments (ITT). Details of the ANLFQ: Satisfaction, BAS-PP, and ANLFQ: Impacts scales have been published.[17]

Key additional efficacy endpoints were achievement of ≥1-grade improvement at maximum contraction over time based on the investigator's and participant's assessments (mITT), and responses of *No sagging or drooping* or *Mild sagging or drooping* among those with *Moderate, Severe,* or *Extreme sagging or drooping* at baseline on the Participant Global Impression of Severity (PGIS)–Jawline at all follow-up visits (ITT and mITT). Efficacy endpoints were assessed bilaterally, requiring participants to achieve the desired outcome on both sides of the neck or jawline. Because the results were comparable in both the ITT and mITT populations, this manuscript focuses on the mITT population and incorporates the ITT data as supplementary information (Supplemental Table 1, located online at https://doi.org/10.1093/asj/sjae220).

Safety evaluations were conducted in the safety population, which included all participants who received the study drug. Adverse events and vital signs parameters (systolic and diastolic blood pressure, heart rate, and temperature) were monitored over time. The safety endpoint was the incidence of adverse events.

Statistical Analysis

A sample of 360 participants from the ITT population and 180 participants from the mITT population was estimated to provide a power >99% for detecting differences in the primary composite endpoint and coprimary endpoints, respectively, between treatment groups.

Table 1. Baseline Demographics and Characteristics

	mITT population		ITT population	
	Placebo (*n* = 195)	OnabotA (*n* = 186)	Placebo (*n* = 217)	OnabotA (*n* = 209)
Age, mean, years (SD)	47.4 (10.11)	48.6 (8.81)	46.9 (10.18)	48.0 (9.57)
Sex, *n* (%)				
Male	9 (4.6)	9 (4.8)	11 (5.1)	11 (5.3)
Female	186 (95.4)	177 (95.2)	206 (94.9)	198 (94.7)
Ethnicity, *n* (%)				
Hispanic or Latino	33 (16.9)	29 (15.6)	35 (16.1)	30 (14.4)
Not Hispanic or Latino	162 (83.1)	157 (84.4)	182 (83.9)	179 (85.6)
Race, *n* (%)				
American Indian or Alaskan Native	0 (0.0)	1 (0.5)	0 (0.0)	1 (0.5)
Asian[a]	7 (3.6)	10 (5.4)	8 (3.7)	11 (5.3)
Black or African American	5 (2.6)	2 (1.1)	5 (2.3)	2 (1.0)
Native Hawaiian or other Pacific Islander	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
White	182 (93.3)	169 (90.9)	202 (93.1)	191 (91.4)
Multiple[b]	1 (0.5)	4 (2.2)	2 (0.9)	4 (1.9)
C-APPS at maximum contraction, *n* (%)				
3 on both sides	41 (21.0)	41 (22.0)	49 (22.6)	44 (21.1)
3 on 1 side, 4 on the other	49 (25.1)	49 (26.3)	54 (24.9)	54 (25.8)
4 on both sides	105 (53.8)	96 (51.6)	114 (52.5)	111 (53.1)
Other	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
P-APPS at maximum contraction, *n* (%)				
3 on both sides	51 (26.2)	40 (21.5)	57 (26.3)	45 (21.5)
3 on 1 side, 4 on the other	53 (27.2)	58 (31.2)	58 (26.7)	65 (31.1)
4 on both sides	88 (45.1)	84 (45.2)	99 (45.6)	95 (45.5)
Other	3 (1.5)	4 (2.2)	3 (1.4)	4 (1.9)

C-APPS, Clinician Allergan Platysma Prominence Scale; ITT, intent-to-treat; mITT, modified intent-to-treat; onabotA, onabotulinumtoxinA; P-APPS, Participant Allergan Platysma Prominence Scale. [a]Asian participants included Japanese, Chinese, Korean, Taiwanese, and Other. [b]Participants who reported multiple races were included in the multiple category only.

Primary and secondary analyses were conducted for the ITT and mITT populations with a prespecified gated hierarchical testing procedure to maintain a family-wise Type I error rate of $\alpha = .05$. The secondary endpoints analyzed over time were excluded from the hierarchical testing.

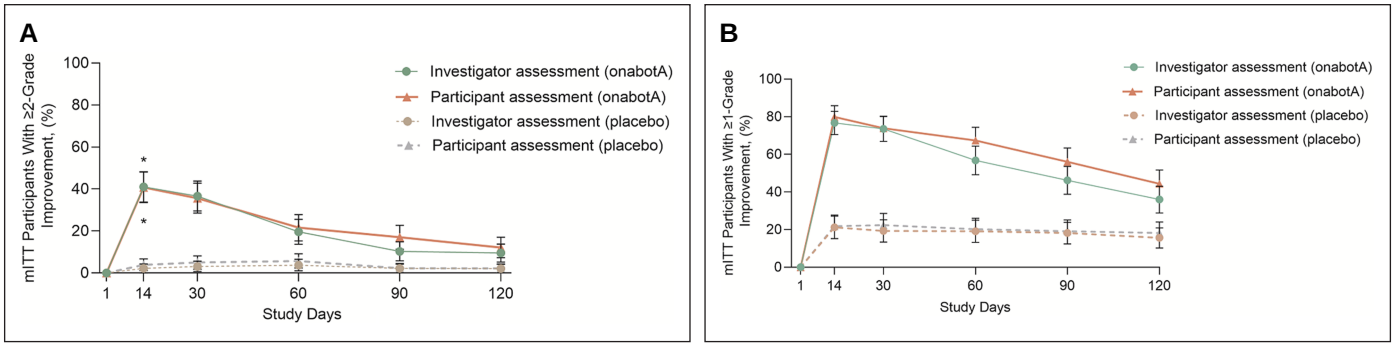


Figure 3. Percentage of participants in the mITT population achieving at least a 2-grade improvement (A) and 1-grade improvement (B) from baseline in platysma prominence severity at maximum contraction, as assessed by investigator and participants, separately. *$P < .0001$ vs placebo; error bars indicate 95% confidence intervals. mITT, modified intent-to-treat; onabotA, onabotulinumtoxinA.

P values were presented for the endpoints that were in the testing hierarchy; in addition, 95% CIs were presented for the responder endpoints.

The primary and secondary endpoints responder analyses were conducted with the Cochran-Mantel-Haenszel (CMH) model, stratified by baseline C-APPS score and investigator site. Continuous variables (ANLFQ: Impacts) were analyzed with analysis of covariance (ANCOVA), with treatment as the main effect, adjusting for baseline C-APPS score and by investigator site, and corresponding baseline patient-reported outcome values as factors. A 2-sided Type I error rate of 0.05 was employed for all calculations. Multiple imputation was utilized for missing data.

RESULTS

Study Participants

Out of the 586 participants screened, 381 were randomized and included in the mITT population (186 to onabotA and 195 to placebo). Of those, 379 received study treatment and 2 did not because they were enrolled in error. A total of 348 (91.3%) mITT participants completed the study. Overall, the main reason for discontinuation from both the onabotA and the placebo groups was withdrawal by participants. No participant was discontinued from the study because of adverse events (Figure 2). The median follow-up time for the mITT participants was 133.2 days.

Demographic and baseline PP characteristics were similar between the onabotA and placebo groups in both the mITT and ITT populations (Table 1 and Supplemental Table 2, located online at https://doi.org/10.1093/asj/sjae220). The mITT population included participants who predominantly self-identified as female (95.3%) and White (92.1%), with a mean age of 48.0 years (age range: 20-71 years) and a mean BMI of 23.3 kg/m^2. All Fitzpatrick skin types were represented, with type II (34.9%) and III (38.3%) being the most common. At baseline, over half of participants (52.8%) were rated as severe PP on both sides based on investigator assessment with the C-APPS, with a similar distribution seen for the P-APPS (45.1%). The majority of participants (52.8%) reported being *A lot bothered* by their vertical neck bands, and 46.7% were *A lot bothered* by the appearance of their jawline. On the PGIS-Jawline, more than 70% of participants in the onabotA group had *Moderate, Severe,* or *Extreme sagging or drooping* (49.5%, 21.5%, 0.5%, respectively). The mean (SD) ANLFQ: Impacts summary score at baseline was 25.5 (3.98).

Efficacy Outcomes

Improvement in Platysma Prominence Severity

OnabotA treatment showed significant improvements in PP from baseline to Day 14 for both coprimary efficacy outcomes. According to the investigator's assessment, 41.0% (95% CI: 33.8%-48.3%) of participants experienced at least a 2-grade improvement at Day 14 compared to 2.2% (95% CI: 0.0%-4.3%, $P < .0001$) in the placebo group (Figure 3A). Similarly, on the participant's self-assessment, 40.8% (95% CI: 33.5%-48.1%) in the onabotA group achieved at least a 2-grade improvement at Day 14 compared to 3.9% (95% CI: 1.1%-6.7%, $P < .0001$) in the placebo group (Figure 3A).

Participants reported that 48.1% (95% CI: 40.7%-55.5%) in the onabotA group achieved Grade 1 (Minimal) or 2 (Mild) at Day 14 compared to 5.2% (95% CI: 1.9%-8.4%, $P < .0001$) in the placebo group. Furthermore, more mITT participants achieved at least a 1-grade improvement in the C-APPS with onabotA than with placebo at Day 14 (onabotA: 76.7%; 95% CI: 70.5%-82.9%; placebo: 21.2%; 95% CI: 15.2%-27.2%), based on the investigator's assessment (Figure 3B). Similar findings for 1-grade improvement were reported by participants (onabotA: 79.9%; 95% CI: 74.0%-85.9%; placebo: 21.8%; 95% CI: 15.7%-27.8%; Figure 3B). Representative photographs of participants before and after treatment on Day 14 are shown in Figure 4.

Satisfaction With the Effect of Treatment

At Day 14, a significantly greater percentage of participants in the onabotA group (61.2%; 95% CI: 54.0%-68.4%) reported being *Satisfied* or *Very satisfied* with the effect of treatment on the ANLFQ: Satisfaction (Follow-up) Item 5 compared to the placebo group (11.8%; 95% CI: 7.0%-16.6%, $P < .0001$) (Figure 5).

Bother From Appearance of Jawline and Vertical Neck Bands

Among participants who reported being at least *Somewhat bothered* by the appearance of their jawline at baseline, 49.4% (95% CI: 42.1%-56.7%) of participants in the onabotA group reported *Not at all bothered* or *A little bothered* on the BAS-PP Item 2 at Day 14 compared to 20.6% (95% CI:14.7%-26.4%) in the placebo group ($P < .0001$) (Figure 6A). On the BAS-PP Item 1 (appearance of vertical neck bands), 47.8% (95% CI: 40.5%-55.1%) of onabotA-treated participants had responses of *Not at all bothered* or *A little bothered* compared to 11.9% (95% CI: 7.2%-16.6%) with placebo (Figure 6B).

Psychosocial Impact of Platysma Prominence

The mean change from baseline in the ANLFQ: Impacts total score was significantly higher ($P < .0001$) in the onabotA group than the placebo group at Day 14 (onabotA, −7.7; placebo, −3.0), Day 30 (onabotA, −8.8; placebo, −3.9), Day 60 (onabotA, −8.2; placebo, −3.5), and Day 90 (onabotA, −7.4; placebo, −3.5). These results suggest that onabotA had a greater positive effect than placebo in improving the psychosocial impact of PP.



Figure 4. Images illustrating the improvement in platysma prominence at maximal contraction of the platysma muscle at baseline and 14 days after treatment with onabotA. (A) A 41-year-old female participant rated independently by both clinician and participant as a Grade 4 (Severe) on both sides at baseline, who achieved a Grade 1 (Minimal) improvement on the left side (B) at Day 14 after receiving onabotA 36 U, with a follow-up of 130 days. (C) A 55-year-old female participant rated Grade 3 (Moderate) on both sides at baseline, as assessed by the investigator and participants, who achieved a Grade 1 improvement on both sides (D) at Day 14 after receiving onabotA 26 U, with a follow-up of 125 days. (E) A 30-year-old female participant with a clinician- and participant-assessed Grade 4 on both sides at baseline, who achieved a Grade 1 improvement on both sides (F) at Day 14. She was monitored for 105 days after receiving onabotA 36 U. The first 2 participants responded *Very satisfied* on the ANLFQ: Satisfaction (Follow-up) Item 5 at Day 14 and improved from being *Somewhat bothered* at Day 1 to *Not at all bothered* on the Bother Assessment Scale–Platysma Prominence (BAS-PP) Item 2 (jawline) and Item 1 (vertical neck bands) at Day 14. The third participant also reported being *Very satisfied* on the ANLFQ: Satisfaction (Follow-up) Item 5 at Day 14, and improved from being *A lot bothered* to *Not at all bothered* on the BAS-PP Items 2 and 1 at Day 14.

Jawline Definition

Among those participants with *Moderate, Severe,* or *Extreme sagging or drooping* at baseline on the PGIS-Jawline, the percentage reporting *No sagging or drooping* or *Mild sagging or drooping* was notably higher in the onabotA group at all time points. At Day 14, the responder rate was 49.8% (95% CI: 40.8%-58.8%) in the onabotA group compared to 15.9% (95% CI: 8.6%-23.2%) in the placebo group (Figure 7). The percentage of responders in the onabotA group increased to 54.4% (95% CI: 45.4%-63.4%) at Day 30 compared to the placebo group (19.3%; 95% CI: 11.6%-27.0%), and although the percentage declined over time, it remained higher than placebo through Day 120.

Safety Outcomes

Treatment-emergent adverse events (TEAEs) were reported by 34 participants out of the 208 who received onabotA (16.3%) and 43 participants out of the 216 who received placebo (19.9%) (Table 2). A total of 8 (3.8%) participants in the onabotA group and 7 (3.2%) in the placebo group reported TEAEs that were considered related to the treatment by the investigator. Treatment-related TEAEs occurring in more than 1% of the study population were: injection site pain (1.0% in the onabotA group, 1.4% in placebo), injection site bruising (1.0% in the onabotA group, 0.5% in placebo), and injection site hemorrhage (1.0% in the onabotA group, 0.5% in the placebo group).

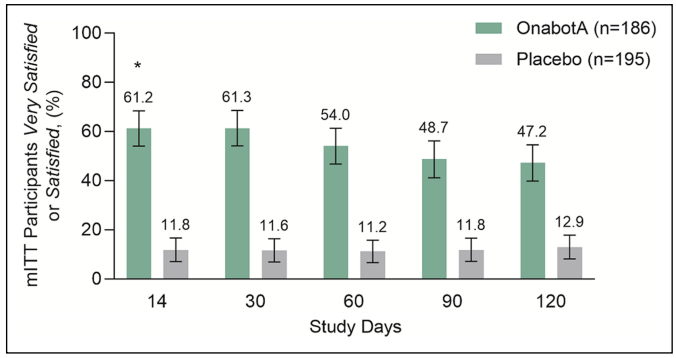


Figure 5. Percentage of participants in the mITT population reporting *Satisfied* or *Very satisfied* on the Appearance of Neck and Lower Face Questionnaire (ANLFQ): Satisfaction (Follow-up) Item 5 (effect of treatment) over time. *$P < .0001$ vs placebo; error bars indicate 95% confidence intervals. mITT, modified intent-to-treat; onabotA, onabotulinumtoxinA.

None of the treatment-related TEAEs were serious, and the majority were localized, mild in severity, generally transient (median duration of 3 days), and resolved spontaneously.

Additional TEAEs were observed but considered unrelated to the study treatment by investigators. Overall, the most frequently reported TEAEs were COVID-19 infection (3.1%), influenza-like illness and nasopharyngitis (1.4% each), and injection site pain (1.2%). The majority of TEAEs were mild (14.2%) or moderate (5.4%) in severity. Two severe TEAEs were reported: 1 participant in the placebo group (0.5%) reported an adnexa uteri cyst event, and 1 participant in the onabotA group (0.5%) reported a COVID-19 infection.

There were no significant changes in vital signs compared with baseline throughout the study. No instances of possible distant spread of toxin events or deaths were reported in the study.

DISCUSSION

Repeated activation of the platysma muscle contributes to the appearance of vertical bands on the neck and blunting of the jawline, making neuromodulation of this muscle a minimally invasive treatment option. In this randomized, placebo-controlled study, we aimed to evaluate the safety and efficacy of onabotA for the temporary improvement of moderate to severe PP with validated measures to assess the impact of treatment, including jawline and neck appearance.[17]

In youth, the elastic deep fascia minimizes appearance of platysma banding during muscle contraction. Decreased fascia elasticity can lead to visible vertical neck bands at rest.[18] Facial movements, such as smiling or talking, can also produce vertical bands that are not visible at rest but become prominent upon mild contraction of the platysma muscle.[11,12] Consistent with other neurotoxin trials, in this study we focused on maximum contraction, when onabotA effects are most evident. No assessments at rest were developed or included in the study.

In this study, we found that treatment with onabotA 26, 31, or 36 U with the described injection paradigm led to significant and meaningful improvements in severity of PP appearance compared to placebo. These improvements met the stringent regulatory criteria established for primary composite and coprimary endpoints (each $P < .0001$), as well as the secondary endpoints (each $P < .0001$). Over 75% of participants in the onabotA group achieved at least a 1-grade improvement from baseline to Day 14 according to the individual assessments of both investigators and participants. Additionally, responder rates

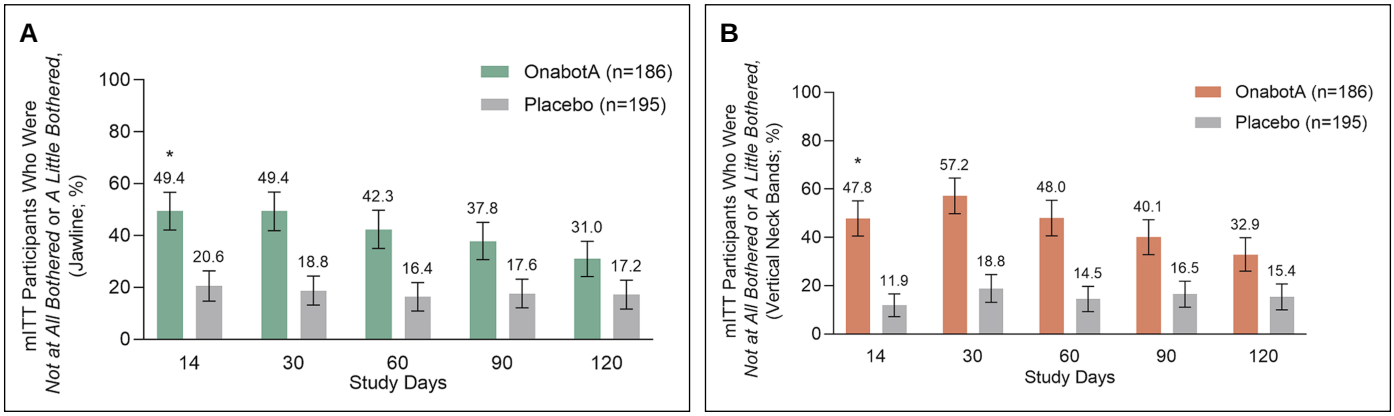


Figure 6. Percentage of participants in the mITT population reporting *Not at all bothered* or *A little bothered* on (A) the Bother Assessment Scale—Platysma Prominence (BAS-PP) Item 2 (jawline) and (B) Item 1 (vertical neck bands) over time. *$P < .0001$ vs placebo; error bars indicate 95% confidence intervals. mITT, modified intent-to-treat; onabotA, onabotulinumtoxinA.

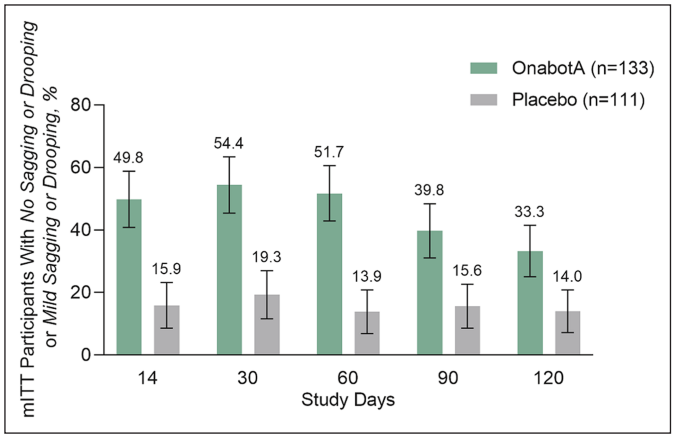


Figure 7. Percentage of participants in the mITT population with responses of *Mild* or *No sagging* on the Participant Global Impression of Severity (PGIS)–Jawline at all follow-up visits, among those who reported *Moderate, Severe,* or *Extreme sagging or drooping* at baseline. Error bars indicate 95% confidence intervals. mITT, modified intent-to-treat; onabotA, onabotulinumtoxinA.

remained consistently higher with onabotA compared to placebo through Day 120, with a gradual decline over time. These findings replicate the other pivotal Phase 3 study and align with previous PP studies that reported onabotA effects lasting between 3 and 4 months.[15,16,19]

The benefits of onabotA treatment for PP were further confirmed by participants, who reported significant improvements from baseline in bother with the appearance of jawline and vertical neck bands, and a reduction in the psychosocial impact related to PP (each $P < .0001$ vs placebo). Treatment satisfaction rates were significantly higher in the onabotA group compared to placebo ($P < .0001$) and were maintained throughout the study. When assessing treatment satisfaction, it is important to consider the unique characteristics of the aging neck and lower face. Other than PP, several factors contribute to the senescence of the neck and lower face, including skin laxity, jowls, submental or subplatysmal lipodystrophy, and horizontal neck bands.[3,6,20] During the consultation visit, it is necessary to evaluate the underlying cause of the patient's aesthetic concerns and establish realistic treatment expectations.[21] Individuals who have good skin elasticity, minimal submental fullness, and no hypertrophic submandibular glands are considered the most suitable onabotA treatment candidates.[10,22,23] Combination therapy with lasers, energy-based devices, and injectable fillers may be recommended to enhance outcomes.[24-27]

Table 2. Summary of Adverse Events in Safety Population

	Placebo ($n = 216$) *n* (%)	OnabotA ($n = 208$) *n* (%)	Total ($n = 424$) *n* (%)
Treatment-emergent adverse events (TEAE)	43 (19.9)	34 (16.3)	77 (18.2)
TEAE related to study treatment	7 (3.2)	8 (3.8)	15 (3.5)
TEAE related to study procedure	6 (2.8)	7 (3.4)	13 (3.1)
TEAE related to study drug	1 (0.5)	2 (1.0)	3 (0.7)
Mild	6 (2.8)	8 (3.8)	14 (3.3)
Moderate	1 (0.5)	0 (0.0)	1 (0.2)
Severe	0 (0.0)	0 (0.0)	0 (0.0)
Treatment-emergent serious adverse events (TESAE)[a]	1 (0.5)	0 (0.0)	1 (0.2)
Related to study treatment	0 (0.0)	0 (0.0)	0 (0.0)
TEAE leading to death	0 (0.0)	0 (0.0)	0 (0.0)
Treatment-emergent adverse events of special interest (AESI)	0 (0.0)	0 (0.0)	0 (0.0)
Possible distant spread of toxin (PDSOT) TEAE	0 (0.0)	0 (0.0)	0 (0.0)
Deaths, *n* (%)	0 (0.0)	0 (0.0)	0 (0.0)

OnabotA, onabotulinumtoxinA. [a]One participant reported a TESAE of adnexa uteri cyst, which was considered not related to the study treatment by the investigator.

Furthermore, participants treated with onabotA reported greater improvements in the definition of the lower face compared to placebo on the PGIS-Jawline. These findings were supported by the significantly greater percentage of onabotA-treated participants who reported being *Not at all bothered* to *A little bothered* by their jawline vs placebo on BAS-PP Item 2 following treatment. The study injection approach specifically targeted the upper section of the platysma muscle, administering 8 U per side approximately 1 to 2 cm below the mandible, to reduce the downward tension exerted by this muscle on the lower face. Previous research has also reported improvements in jawline contour with onabotA treatment of the platysma.[19,23,28,29]

The onabotA doses used in this study were selected to optimize tolerability while maintaining efficacy, as determined by a previous phase 2 dose-escalating study.[15] Although the phase 2 study reported a trend of greater efficacy with higher onabotA doses, these higher doses may also increase the incidence of treatment-related adverse events, such as dysphagia.[12,15,23,30,31] Apart from selecting an appropriate dose, the injection paradigm is crucial to minimize risk of complications. The injection paradigm in this study was tailored to the participant's platysma prominence presentation at baseline. Treatment inferior to the mandible relaxes the muscle sheet at and below the jawline, while treatment to continuous neck bands minimizes or eliminates their appearance with contraction. With this paradigm, an onabotA dose of 1 U per injection was given in the neck area compared to 2 U per injection below the jawline, or 4 U per injection for upper facial lines.[32] The dose to the anterior neck (the part of the neck over the deglutition muscles) was limited to 10 U, if 2 anterior bands were treated, 1 on each side. Participants with only 1 anterior neck band on 1 side (and no band on the other side) received a total of 5 U in the anterior neck. This helped to minimize risks of toxin effect on the underlying muscles of the anterior neck and reduce the likelihood of negative outcomes (eg, dysphagia). Importantly, participants with Grade 3 on both sides at baseline (1 band treated on each side) might not have received treatment in the anterior neck, because single bands are usually located in the middle rather than in the anterior or posterior neck. Additionally, due to the thin and superficial nature of the platysma muscle, it is important that all injections are administered superficially into the muscle.[18] Understanding the anatomy of the area and using a proper injection technique are necessary to achieve optimal outcomes.

OnabotA was well tolerated in this study, with similar incidence rates of TEAEs and treatment-related TEAEs between the onabotA and placebo groups. Most TEAEs were considered mild and unrelated to the study treatment, according to the investigator's evaluation. The incidence of adverse events was similar to that of the other pivotal Phase 3 study, with no reports of dysphagia or neck muscular weakness. No evidence of possible distant spread of toxin or diffusion was identified.

This analysis has some limitations. Despite the geographic range of the study sites, the study population primarily consisted of female participants and individuals with lighter skin types. Additionally, this analysis only addressed treatment outcomes after a single onabotA treatment. The effects of up to 3 onabotA treatments for PP have been evaluated in an 8-month open-label extension study. Taken together, these studies provide additional insight into the long-term safety and efficacy of repeated onabotA treatment.

CONCLUSIONS

OnabotA treatment with 26, 31, and 36 U was well tolerated and effective in reducing the severity of platysma prominence, with statistically significant and clinically meaningful improvements compared to placebo. Participants treated with onabotA reported higher treatment satisfaction, with reduced bother and improved psychosocial impact related to PP, compared to those who received placebo. Furthermore, the onabotA treatment approach was effective in improving jawline definition.

Supplemental Material

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Disclosures

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